

Calibration using FEM modeling:

1. Data acquisition:

- Select N_p points $\mathbf{p}_i$ (3×1) for $i = 1, 2, \dots, N_p$ in the work space.
- Perform N_I Finite Element Method (FEM) simulations using the input current vectors $\mathbf{I}_j = [I_{j1} \ I_{j2} \ I_{j3} \ I_{j4} \ I_{j5} \ I_{j6}]^T$ for $j = 1, 2, \dots, N_I$, ensuring that the combined current matrix $\mathbf{F} = [\mathbf{I}_1 \ \mathbf{I}_2 \ \dots \ \mathbf{I}_{N_I}]$ has a rank of six. For each simulation.
- Record the magnetic flux density $\mathbf{b}_{ij} = \mathbf{b}^{FEM}(\mathbf{p}_i, \mathbf{I}_j)$ (3×1) for $i = 1, 2, \dots, N_p$.
- Record the sensor voltage v_{jk} for $k = 1, 2, \dots, 6$ and for $j = 1, 2, \dots, N_I$.

2. Modeling hexapole charge location:

- Actuator coordinate system: $[\hat{\mathbf{u}} \ \hat{\mathbf{v}} \ \hat{\mathbf{w}}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
- One-parameter model: Introduce a characteristic length scale $\hat{\ell}$, relating the geometry of the hexapole tweezers to the discrete locations of the six magnetic charges in the lumped-parameter model.

$$\bar{\mathbf{p}}_c = [\bar{\mathbf{p}}_{c1} \ \bar{\mathbf{p}}_{c2} \ \bar{\mathbf{p}}_{c3} \ \bar{\mathbf{p}}_{c4} \ \bar{\mathbf{p}}_{c5} \ \bar{\mathbf{p}}_{c6}] = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

$$\mathbf{p}_c = \hat{\ell} \cdot \bar{\mathbf{p}}_c$$

- 18-parameter model: Introduce a characteristic length scale $\hat{\ell}$ and a 1×17 bias vector $\mathbf{e}^T$, relating the geometry of the hexapole tweezers to the discrete locations of the six magnetic charges in the lumped-parameter model.

$$\mathbf{e}^T = \left[\underbrace{e_{1x} \ e_{1y} \ e_{1z}}_{\mathbf{e}_1^T} \ \underbrace{e_{2x} \ e_{2y} \ e_{2z}}_{\mathbf{e}_2^T} \ \underbrace{e_{3x} \ e_{3y} \ e_{3z}}_{\mathbf{e}_3^T} \ \underbrace{e_{4x} \ e_{4y} \ e_{4z}}_{\mathbf{e}_4^T} \ \underbrace{e_{5x} \ e_{5y} \ e_{5z}}_{\mathbf{e}_5^T} \ \underbrace{e_{6x} \ e_{6y}}_{\mathbf{e}_6^T} \right]$$

$$e_{6z} = e_{1x} - e_{2x} + e_{3y} - e_{4y} + e_{5z}$$

$$\mathbf{e}_6^T = [e_{6x} \ e_{6y} \ (e_{1x} - e_{2x} + e_{3y} - e_{4y} + e_{5z})]$$

$$\begin{aligned}\bar{\mathbf{p}}_c(\mathbf{e}) &= [\hat{\mathbf{u}} \quad -\hat{\mathbf{u}} \quad \hat{\mathbf{v}} \quad -\hat{\mathbf{v}} \quad \hat{\mathbf{w}} \quad -\hat{\mathbf{w}}] + [\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3 \quad \mathbf{e}_4 \quad \mathbf{e}_5 \quad \mathbf{e}_6] \\ &= \begin{bmatrix} 1 + e_{1x} & -1 + e_{2x} & e_{3x} & e_{4x} & e_{5x} & e_{6x} \\ e_{1y} & e_{2y} & 1 + e_{3y} & -1 + e_{4y} & e_{5y} & e_{6y} \\ e_{1z} & e_{2z} & e_{3z} & e_{4z} & 1 + e_{5z} & -1 + e_{6z} \end{bmatrix}\end{aligned}$$

$$\mathbf{p}_c = \hat{\ell} \cdot \bar{\mathbf{p}}_c$$

3. Nondimensionalized 3×6 spatial function matrix $\mathbf{S}(\bar{\mathbf{p}}_i)$ for each point.

a. One-parameter model:

$$\bar{\mathbf{p}}_i = \mathbf{p}_i / \hat{\ell} \text{ for } i = 1, 2, \dots, N_p.$$

$$\mathbf{S}(\mathbf{p}_i / \hat{\ell}) = \begin{bmatrix} \frac{\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_{c1}}{\|\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_{c1}\|^3} & \frac{\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_{c2}}{\|\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_{c2}\|^3} & \dots & \frac{\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_{c6}}{\|\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_{c6}\|^3} \end{bmatrix} (3 \times 6) \text{ for } i = 1, 2, \dots, N_p.$$

b. 18-parameter model:

$$\bar{\mathbf{p}}_i = \mathbf{p}_i / \hat{\ell} \text{ for } i = 1, 2, \dots, N_p.$$

$$\mathbf{S}(\mathbf{p}_i / \hat{\ell}, \hat{\mathbf{e}}) = \begin{bmatrix} \frac{\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_{c1}}{\|\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_{c1}\|^3} & \frac{\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_{c2}}{\|\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_{c2}\|^3} & \dots & \frac{\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_{c6}}{\|\bar{\mathbf{p}}_i - \bar{\mathbf{p}}_{c6}\|^3} \end{bmatrix} (3 \times 6) \text{ for } i = 1, 2, \dots, N_p.$$

4. Cost function:

a. One-parameter model:

$$J(\hat{\ell}) = \sum_{j=1}^{N_I} \left\{ \sum_{i=1}^{N_p} \mathbf{b}_{ij}^T \mathbf{b}_{ij} - \sum_{i=1}^{N_p} \mathbf{b}_{ij}^T \mathbf{S}(\mathbf{p}_i / \hat{\ell}) \left(\sum_{i=1}^{N_p} \mathbf{S}^T(\mathbf{p}_i / \hat{\ell}) \mathbf{S}(\mathbf{p}_i / \hat{\ell}) \right)^{-1} \sum_{i=1}^{N_p} \mathbf{S}^T(\mathbf{p}_i / \hat{\ell}) \mathbf{b}_{ij} \right\}$$

b. 18-parameter model:

$$\begin{aligned}J(\hat{\ell}, \hat{\mathbf{e}}) &= \sum_{j=1}^{N_I} \left\{ \sum_{i=1}^{N_p} \mathbf{b}_{ij}^T \mathbf{b}_{ij} \right. \\ &\quad \left. - \sum_{i=1}^{N_p} \mathbf{b}_{ij}^T \mathbf{S}(\mathbf{p}_i / \hat{\ell}, \hat{\mathbf{e}}) \left(\sum_{i=1}^{N_p} \mathbf{S}^T(\mathbf{p}_i / \hat{\ell}, \hat{\mathbf{e}}) \mathbf{S}(\mathbf{p}_i / \hat{\ell}, \hat{\mathbf{e}}) \right)^{-1} \sum_{i=1}^{N_p} \mathbf{S}^T(\mathbf{p}_i / \hat{\ell}, \hat{\mathbf{e}}) \mathbf{b}_{ij} \right\}\end{aligned}$$

5. Minimize cost function:

a. One-parameter model: Minimize $J(\hat{\ell}) \Rightarrow \hat{\ell}$ (Initial guess: $\hat{\ell} = \ell_{design}$)

b. 18-parameter model: Minimize $J(\hat{\ell}, \hat{\mathbf{e}}) \Rightarrow \hat{\ell}, \hat{\mathbf{e}}$ (Initial guess: $\hat{\ell} = \ell_{design}$ and $\hat{\mathbf{e}} = \mathbf{0}$)

6. Calculate $\mathbf{g}_j = \left(\sum_{i=1}^{N_p} \mathbf{S}^T(\bar{\mathbf{p}}_i) \mathbf{S}(\bar{\mathbf{p}}_i) \right)^{-1} \left(\sum_{i=1}^{N_p} \mathbf{S}^T(\bar{\mathbf{p}}_i) \mathbf{b}_{ij} \right) (6 \times 1)$ for $j = 1, 2, \dots, N_I$.

7. Define a $6 \times N_I$ matrix:

$$\mathbf{G} = [\mathbf{g}_1 \quad \mathbf{g}_2 \quad \dots \quad \mathbf{g}_{N_I}]$$

8. Determine $\hat{g}_B \bar{\mathbf{K}}_I$: $\hat{g}_B \bar{\mathbf{K}}_I = \mathbf{H} = \mathbf{G} \mathbf{F}^T (\mathbf{F} \mathbf{F}^T)^{-1}$

$$\hat{g}_B = \frac{6}{5} h_{11} \text{ and } \bar{\mathbf{K}}_I = \frac{5}{6 \cdot g_{11}} \mathbf{H}$$